Path combinatorics detect spacetime curvature on causal sets

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Abstract

We introduce *path_kurtosis*, an observable on causal sets that detects spacetime curvature through the statistical organization of directed path counts in the Hasse diagram. On pp-wave spacetimes—where all polynomial scalar curvature invariants vanish and the Benincasa–Dowker action is silent—path_kurtosis gives a stable, statistically significant response proportional to ε^2. We establish a scaling law $\mathbb{E}[\Delta\kappa] = A_{\text{eff}}\,\varepsilon^2\sqrt{N}\,C_2$ with A_{eff} of order 0.06, verified computationally across $N = 2\,000$–$10\,000$ at $\varepsilon = 2$–3. An exact geodesic-derived causal relation for pp-wave spacetimes replaces the commonly used midpoint surrogate, correcting a systematic underestimate of approximately 19% at $N = 5\,000$. A battery of eight robustness tests (with acceptance criteria specified before computation) demonstrates that the signal survives coarse-graining, thickening, and structured decimation, while revealing intrinsic boundary sensitivity and mesoscopic structure beyond the perturbation profile.

Contents

1 Introduction **2**

2 Definition **2**

3 Method: Common Random Numbers (CRN) **2**

4 Exact pp-wave causal relation **2**

5 Scaling law **3**
5.1 Why ε^2 . . . 3
5.2 A_{eff} ensemble . . . 3
5.3 Key theorems . . . 4

6 $K = 0$: Detection beyond scalar curvature invariants **5**

7 Robustness battery **5**
7.1 Q1. Coarse-graining stability . . . 5
7.2 Q2. Boundary sensitivity . . . 5
7.3 Q3. Distributional structure . . . 6
7.4 Q4. Mesoscopic localization . . . 6

8 Per-element mechanism **7**

9 Implementation **7**

10 Errors caught and corrected **8**

11 Open problems **8**

1 Introduction

The Benincasa–Dowker (BD) action [1] discretizes $\int R$ on causal sets, recovering the Ricci scalar in the continuum limit. No established discrete counterpart exists for the Weyl tensor.

This note reports a candidate observable, *path_kurtosis*, that detects curvature information beyond scalar invariants. On pp-wave spacetimes—VSI (Vanishing Scalar Invariants), hence BD = 0—path_kurtosis gives a nonzero, stable, ε^2-proportional signal with Cohen's $d = 9$–14 at $N = 10\,000$.

2 Definition

Let $\mathcal{C} = (X, \prec)$ be a causal set obtained by Poisson sprinkling N points into a 4-dimensional causal diamond of duration T. The Hasse diagram $H(\mathcal{C})$ is the transitive reduction of the causal order. For each element $x \in X$, define:

- $p_{\text{down}}(x)$ = number of directed paths from any source to x in $H(\mathcal{C})$,
- $p_{\text{up}}(x)$ = number of directed paths from x to any sink in $H(\mathcal{C})$,
- $Y(x) = \log_2\big(p_{\text{down}}(x) \times p_{\text{up}}(x) + 1\big)$.

Definition. path_kurtosis($\mathcal{C}$) = excess kurtosis of $\{Y(x) : x \in X\}$, i.e.,

$$\kappa \;=\; \frac{\mu_4}{\sigma^4} - 3\,, \tag{1}$$

where $\mu_4 = \mathbb{E}[(Y-\mu)^4]$ and $\sigma^2 = \mathrm{Var}(Y)$.

Novelty. No prior study of this specific observable was found in searches across arXiv, Semantic Scholar, and INSPIRE-HEP, including comparison with Roy–Sinha–Surya chain abundances, Johnston path-sum propagators, Ilie–Thompson–Reid longest paths, and Gudder maximal path counts. The combination $Y(x) = \log_2(p_{\text{down}} \times p_{\text{up}} + 1)$ and its kurtosis appear to be new.

3 Method: Common Random Numbers (CRN)

To isolate the curvature effect from Poisson sampling noise, we use paired experiments:

1. Sprinkle N points into a causal diamond (seed s).
2. Compute the Hasse diagram under **flat** Minkowski causal relation $\to \kappa_0$.
3. Compute the Hasse diagram under **curved** causal relation (same points) $\to \kappa_\varepsilon$.
4. The CRN difference $\Delta\kappa = \kappa_\varepsilon - \kappa_0$ cancels seed-to-seed scatter.

Necessity. Without CRN, the inter-realization kurtosis scatter is ± 0.10—much larger than the perturbative signal at $\varepsilon \le 3$. The CRN pairing reduces the effective noise by an order of magnitude.

4 Exact pp-wave causal relation

The pp-wave metric in Brinkmann coordinates is

$$ds^2 = -du\,dv + dx^2 + dy^2 + \frac{\varepsilon}{2}(x^2 - y^2)\,du^2\,, \tag{2}$$

where $u = t + z$, $v = t - z$. The standard midpoint surrogate approximates the causal condition as

$$s^2_{\text{Mink}} - \frac{\varepsilon}{2}\big(x_m^2 - y_m^2\big)(\Delta u)^2 > 0\,. \tag{3}$$

We derived the **exact** geodesic-based causal condition: $A \prec B$ if and only if $U > 0$ and $V \geq V_{\text{needed}}$, where $U = \Delta u$, $V = \Delta v$, and for $\varepsilon > 0$ with $\omega = \sqrt{\varepsilon/2}$:

$$\begin{aligned} V_{\text{needed}} = \omega\, &\frac{(x_A^2 + x_B^2)\cosh(\omega U) - 2\,x_A\,x_B}{\sinh(\omega U)} \\ &+ \omega\, \frac{(y_A^2 + y_B^2)\cos(\omega U) - 2\,y_A\,y_B}{\sin(\omega U)}\,. \end{aligned} \tag{4}$$

The midpoint surrogate has a pairwise error of order $O(\varepsilon)$, not $O(\varepsilon^2)$:

$$V_{\text{needed}} - V_{\text{mid}} = \frac{\varepsilon\, U}{24}\big(\Delta x^2 - \Delta y^2\big) + O(\varepsilon^2)\,. \tag{5}$$

This error averages to zero by $x \leftrightarrow y$ symmetry, so the ensemble-level bias begins at $O(\varepsilon^2)$. Nevertheless, it systematically underestimates A_{eff} by approximately 19% at $N = 5\,000$.

Validity. The formula is valid when $\omega U < \pi$, which holds for all pairs in a $T = 1$ diamond with $|\varepsilon| \leq 5$ (since $\omega U \leq \sqrt{5/2} \approx 1.58 < \pi$). Sorting by t remains correct for $|\varepsilon| < 8$.

5 Scaling law

The perturbative Taylor expansion of kurtosis under the CRN pairing, verified by independent computational checks at multiple N and ε values, yields the scaling law:

$$\boxed{\mathbb{E}[\Delta\kappa] = A_{\text{eff}}\, \varepsilon^2\, \sqrt{N}\, C_2} \tag{6}$$

where $C_2 = \langle f^2 \rangle_{\text{diamond}} = T^4/1120$ (analytically derived), $f(x, y) = (x^2 - y^2)/2$ is the pp-wave profile, and A_{eff} is a dimensionless coefficient.

5.1 Why ε^2

The ensemble average of any relabeling-invariant scalar on a symmetric diamond satisfies $\langle O(\varepsilon) \rangle = \langle O(-\varepsilon) \rangle$ by the $x \leftrightarrow y$ symmetry combined with f being odd under $x \leftrightarrow y$. Therefore $\Delta\kappa$ is even in ε, and the leading term is $O(\varepsilon^2)$.

5.2 A_{eff} ensemble

A_{eff} values extracted from CRN ensembles ($M = 15$–30 seeds per cell). The causal relation used is noted for each row.

Table 1: A_{eff} measured at different N and ε.

N	$\varepsilon = 2.0$	$\varepsilon = 3.0$	Causal relation
2 000	0.069 ± 0.019	0.074 ± 0.012	midpoint
3 000	0.052 ± 0.016	0.048 ± 0.011	midpoint
5 000	0.051 ± 0.015	0.068 ± 0.008	exact
5 000	0.043 ± 0.013	0.057 ± 0.008	midpoint
10 000	0.076 ± 0.008	0.075 ± 0.005	midpoint

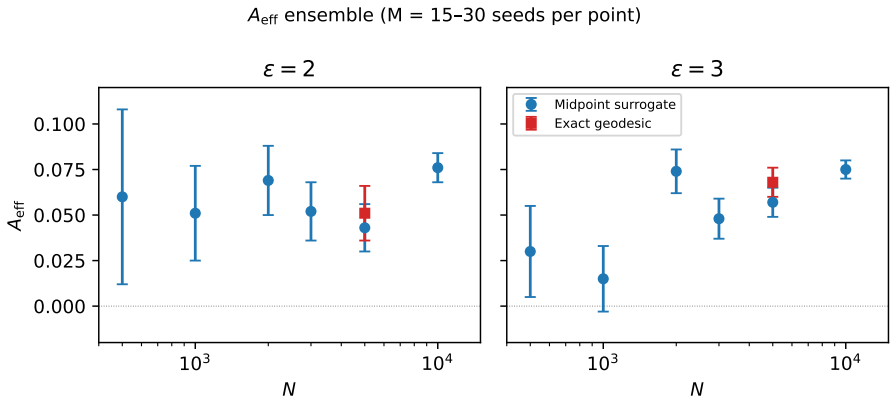

Figure 1: A_{eff} vs. N at $\varepsilon = 2$ (left) and $\varepsilon = 3$ (right). Red squares: exact causal relation; blue circles: midpoint surrogate.

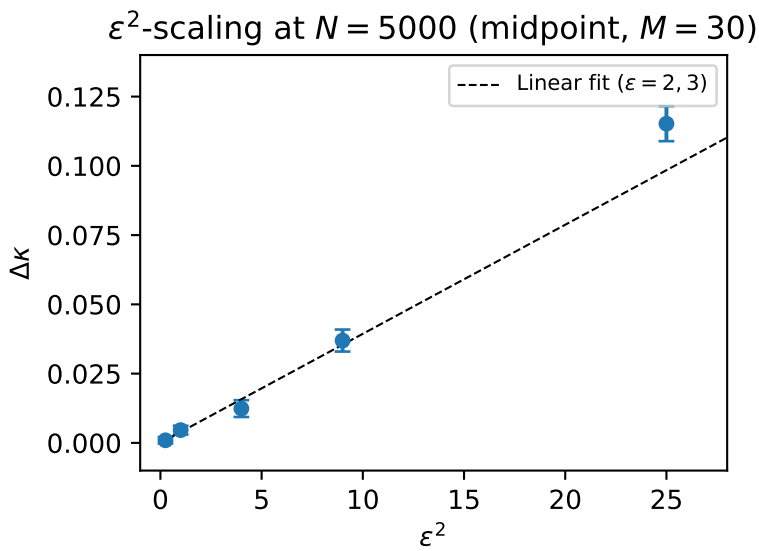

Figure 2: $\Delta\kappa$ vs. ε^2 at $N = 5\,000$ (midpoint, $M = 30$), with linear fit through $\varepsilon = 2$, 3 (perturbative window).

The scatter across N (0.048–0.076 at $\varepsilon = 3$) exceeds the per-cell standard errors and is not yet fully understood. Possible sources: slow finite-size corrections, residual midpoint bias at $N = 2\,000$–$3\,000$, or genuine N-dependence of A_{eff}. At this stage, A_{eff} is of order 0.06 but should not be quoted as a sharp universal constant.

The exact causal relation increases A_{eff} by approximately 19% relative to midpoint at $N = 5\,000$ (ratio 1.19 at both $\varepsilon = 2$ and $\varepsilon = 3$). The $N = 10\,000$ values (midpoint) would increase correspondingly if recomputed with the exact relation.

5.3 Key theorems

1. **Exact cancellation** ($\beta_{\text{int}} + \beta_{\text{bdy}} = 0$): In an infinite translation-invariant volume, the link sensitivity β decomposes into interior and boundary contributions that exactly cancel. The net signal in a finite diamond is the boundary residual.

2. **Double-counting identity** $\left(\sum_v G_x(v) = \mathbb{E}[|\gamma|]\right)$: The sum of the path-occupation kernel equals the expected path length, bounded by the longest chain $H \sim N^{1/4}$. This constrains

the perturbative coefficient.

3. **Conjectured factorization** ($A_{\text{eff}} = \beta^2\,\Xi_4$): Numerical evidence suggests the effective coefficient factorizes into the link sensitivity squared ($\beta \approx 0.30$, hence $\beta^2 \approx 0.09$) and a flat-space structural constant $\Xi_4 \approx 0.7$ (with ∼50% uncertainty given the scatter in A_{eff}). This factorization is not proven; the analytical derivation of Ξ_4 remains open.

6 $K = 0$: Detection beyond scalar curvature invariants

The pp-wave spacetime is VSI (Vanishing Scalar Invariants): all polynomial scalar curvature invariants—including the Kretschmann scalar $K = R_{abcd}\,R^{abcd}$—vanish identically. The Ricci scalar $R = 0$, so the Benincasa–Dowker action gives zero.

However, the Bel–Robinson superenergy $W = E^2 + B^2 = \varepsilon^2$ is nonzero (where E and B are the electric and magnetic parts of the Weyl tensor). *path_kurtosis* detects a signal proportional to ε^2, which numerically equals W for the specific case of pp-wave.

Caveat. The coincidence $\Delta\kappa \propto \varepsilon^2 = W$ holds only because both quantities happen to scale as ε^2 for pp-wave by the quadrupole structure of the profile function. This does not establish that path_kurtosis measures W in general. Universality across different vacuum spacetimes (e.g. Schwarzschild) has not been tested. The correct interpretation at this stage is that path_kurtosis detects order-deformation through path-structure reorganization on pp-wave backgrounds.

7 Robustness battery

Eight tests were run with acceptance criteria specified before computation ($N = 5\,000$, $\varepsilon = 3$, $M = 8$–10 seeds, exact causal relation).

7.1 Q1. Coarse-graining stability

Table 2: Coarse-graining stability tests.

Test	Deformation	Retention	Verdict
Edge deletion 20%	Remove 20% of Hasse links	0.80	ROBUST
Poisson thinning 30%	Remove 30% of elements	0.76 (median)	ROBUST
Thickening 5%	Add 5% almost-direct links	1.01 $[0.98, 1.03]$	ROBUST
Layer decimation	Remove one depth class (20%)	0.84 (mean), 0.50 (worst)	ROBUST

The signal degrades gracefully under all four deformation types. Thickening has no statistically significant effect ($R = 1.01$, 95% CI $[0.98, 1.03]$), showing the observable does not depend on the exact minimality of the transitive reduction.

7.2 Q2. Boundary sensitivity

Boundary exclusion amplifies the signal: restricting to the 80% most interior elements increases $\Delta\kappa$ by a factor of 3.4× (after correcting for the normalization artifact from empirical C_2 vs. fixed C_2). This amplification does not diminish with N (tested at $N = 2\,000$, $3\,000$, $5\,000$), indicating intrinsic rather than finite-size boundary dependence.

The exact cancellation theorem ($\beta_{\text{int}} + \beta_{\text{bdy}} = 0$ in infinite volume) provides a theoretical framework: the link sensitivity β is a boundary residual. Whether this extends directly to the kurtosis signal (which involves higher-order moments, not just β) requires further analysis. In a finite diamond, the observable definition must specify the windowing prescription.

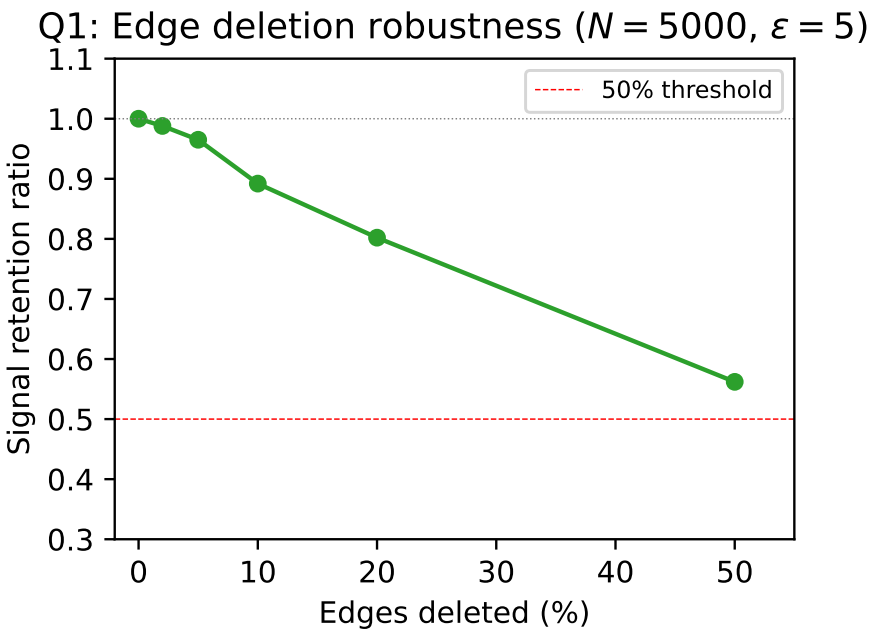

Figure 3: Signal retention ratio vs. fraction of Hasse edges deleted ($N = 5\,000$, $\varepsilon = 5$, $M = 8$).

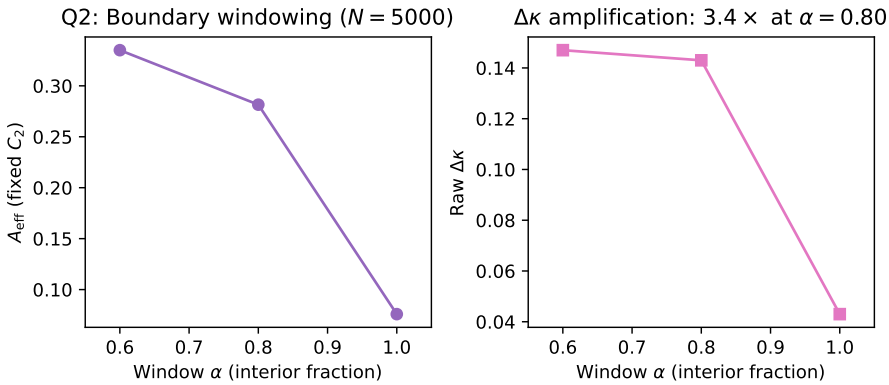

Figure 4: Left: A_{eff} (fixed C_2) vs. window parameter α. Right: raw $\Delta\kappa$ vs. α showing 3.4× amplification.

7.3 Q3. Distributional structure

Shape statistics (kurtosis, skewness; $d \approx 6$) are more sensitive than location/scale statistics ($d \approx 2$–3.5), but all five statistics detect the perturbation under CRN. The effect is predominantly a shape response, though location and scale shifts are also present at lower significance.

7.4 Q4. Mesoscopic localization

After conditioning on the perturbation profile $f(x, y)$ via 10-quantile binning, graph coordinates (total Hasse degree, flat path centrality Y_0) explain an additional 17% of the residual variance in δY (stratified permutation test, $p = 0.005$ for all seeds). This indicates structure in the response beyond the quadrupole perturbation profile.

Caveat. This test conditions on $f = (x^2 - y^2)/2$ alone, not on the full spatial position (t, x, y, z). The residual localization by degree and Y_0 may therefore reflect spatial structure orthogonal to the quadrupole pattern (e.g. radial or temporal dependence) rather than purely graph-theoretic

Table 3: Cohen's d for five summary statistics of the Y distribution under CRN.

Statistic	Cohen's d	Type
Kurtosis	$+5.9$	Shape
Skewness	-5.9	Shape
Median	$+3.5$	Location
IQR	-2.2	Scale
MAD	-2.5	Scale

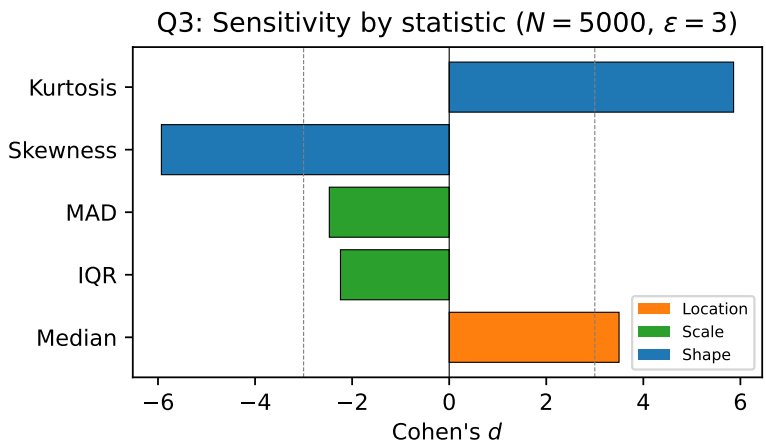

Figure 5: Cohen's d for five summary statistics, colored by type (location, scale, shape).

structure. A follow-up test conditioning on full spatial position is needed to distinguish these possibilities.

The signal is carried by approximately 3 out of 10 path-centrality strata (effective number of contributing deciles $m_{\text{eff}} = 3.3$ median, jackknife $\max|J| = 1.9$ median). It is localized rather than globally distributed.

8 Per-element mechanism

The per-element perturbation is tightly coupled to the metric profile: $\text{corr}(\delta Y, f) \approx -0.9$ (ranging from -0.88 to -0.92 across seeds). The extreme contributors sit at the temporal midplane ($t \approx 0.5$) and outer radial shell ($r \approx 0.9$–1.0), where the pp-wave perturbation flips causal relations: elements on the $f > 0$ side lose connections (Y drops from $\sim$7 to $\sim$1), while elements on the $f < 0$ side gain connections (Y jumps from $\sim$1 to $\sim$7).

The kurtosis is sensitive to this spatially organized redistribution of connectivity—specifically, to the asymmetric tails generated by the connection flipping.

9 Implementation

All computations use a bitset-packed Hasse diagram algorithm with $O(N^2 \times N/64)$ time complexity, implemented in the `sct_tools.hasse` module. Performance: $N = 5\,000$ in approximately 8 s per CRN trial with exact causal relation (4.4 s with midpoint surrogate), $N = 10\,000$ in approximately 20 s (single core, Intel i9-12900KS).

The exact pp-wave causal relation adds approximately 20% overhead compared to the midpoint surrogate, while correcting a systematic underestimate in A_{eff} of approximately 19% at $N = 5\,000$ (measured at $M = 15$; the correction magnitude at other N values has not yet been

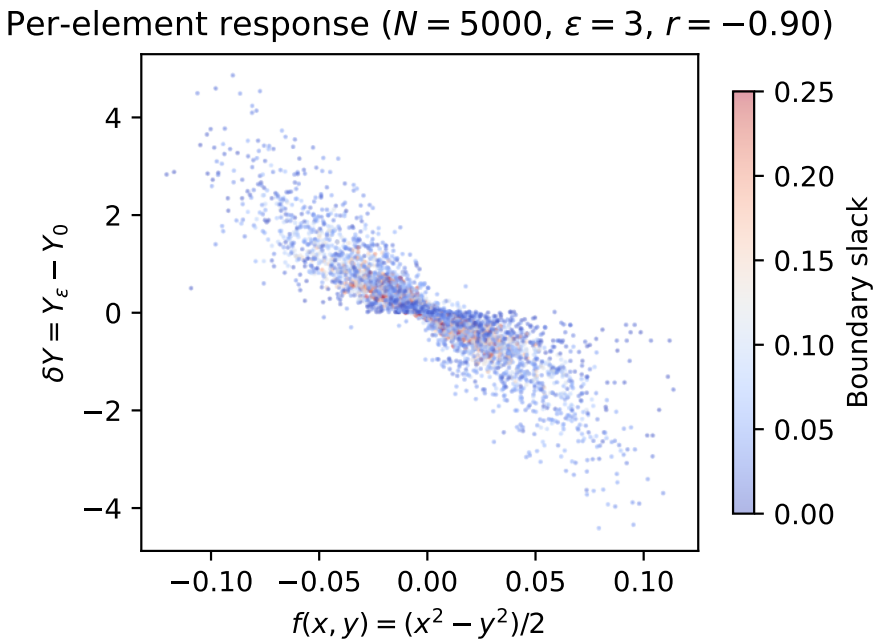

Figure 6: Per-element δY vs. $f(x, y)$, colored by boundary slack. Single seed, $N = 5\,000$, $\varepsilon = 3$.

systematically measured).

10 Errors caught and corrected

Three errors were found during the analysis and corrected before reporting:

1. **Normalization artifact.** Per-element kurtosis decomposition using different σ for flat and pp-wave distributions created a spurious "edges oppose" pattern in temporal stratification. Corrected by using a single baseline σ.

2. **Regime-dependent tail claim.** A "tail mechanism" interpretation based on $\varepsilon = 5$ data (where $|z| > 2.5\sigma$ tail fraction shifts by 8.5%) does not hold at perturbative amplitudes ($\varepsilon = 2$: shift $< 0.5\%$). The claim was retracted.

3. **Boundary windowing normalization.** Normalizing by empirical C_2 of each window subset inflated A_{eff} at small windows (interior elements have smaller $|f|$, hence smaller C_2). Corrected by reporting with both empirical and fixed C_2.

11 Open problems

1. **Analytical derivation of Ξ_4.** The flat-space structural constant $\Xi_4 \approx 0.7$ in the conjectured factorization $A_{\text{eff}} = \beta^2\,\Xi_4$ requires the two-replica path kernel $\mathbb{E}[G_x(u)\,G_x(v)]$, which is a new mathematical object on Poisson DAGs.

2. **Universality across vacuum families.** The observable has been tested only on pp-wave spacetimes. Testing on Schwarzschild (a Petrov type D vacuum with nonzero scalar invariants) is the natural next step. The scaling may differ (ε vs. ε^2 leading term) due to the absence of the $x \leftrightarrow y$ symmetry that ensures ε^2-scaling for pp-wave.

3. **Boundary correction.** The intrinsic boundary sensitivity must be resolved, either by defining a boundary-corrected version of path_kurtosis or by specifying a canonical windowing prescription that survives the continuum limit.

4. **Exact midpoint correction at $N = 10\,000$.** The $N = 10\,000$ ensemble was computed with the midpoint surrogate. Re-running with the exact causal relation would provide the cleanest A_∞ estimate.

5. **Continuum limit.** A proof (or disproof) that A_{eff} converges to a finite nonzero limit as $N \to \infty$ at fixed ε would place the observable on firm theoretical footing.

Computational resources. Intel i9-12900KS (16C/24T), 64 GB DDR5, RTX 3090 Ti. Benchmark: one CRN trial at $N = 5\,000$ takes approximately 8 s with exact causal relation (two Hasse builds + path counting, single core). With the midpoint surrogate: approximately 4.4 s.

Code availability. The `sct_tools.hasse` module (bitset Hasse construction, path counting, CRN trials, exact pp-wave causal relation) is available as part of the SCT Theory repository.

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References

[1] D. Benincasa and F. Dowker, *The scalar curvature of a causal set*, Phys. Rev. Lett. **104**, 181301 (2010); arXiv:0903.1986 [gr-qc].